Bayesian calibration of a Cahn-Hilliard surrogate model

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Abstract

Block copolymer thin films spontaneously assemble into nanoscale structures and is of interest in semiconductor and battery manufacturing. [3] Within the Multifaceted Mathematics for Predictive Digital Twins (M2DT) collaboration to develop a hierarchy of digital twins, this project focuses on developing the Bayesian calibration framework for a physics-informed surrogate model of self-assembling block copolymers. Molecular Dynamics (MD) simulations capture this process with detailed molecular physics, but are computationally expensive and slow, posing challenges to large parameter studies and design optimization. A system identification problem is posed, to find the parameters based on the system we want to model. The Cahn-Hilliard free energy functional describes the spontaneous self-assembly of block copolymers, but the parameters of the model are unknown. We aim to establish a Bayesian calibration workflow to find plausible regions of a surrogate model's parameter space given observation data and quantify uncertainty over these parameters that arise from limited calibration data and model discrepancy. A semi-implicit spectral solver is used to evaluate the Cahn-Hilliard partial differential equation at different points in parameter space. Four Fourier-based scalar quantities of interest (QoIs) are extracted from the 3D concentration fields produced by both the MD and Cahn-Hilliard models to compute residuals in the likelihood function. A gaussian process emulator is trained on the mapping from model parameters to QoIs, eliminating the need to repeatedly solve the PDE during calibration. The posterior distributions are sampled usig markov Chain Monte Carlo with a No-U-Turn (NUTS) sampler. Joint conditioning on the three QoIs produced a broadly consistent posterior distirbutions over plausible parameter regions.

1 Introduction

Digital twins are computational representations of physical systems that combine physics-based models with experimental or simulation data with the goal of Optimal Experiment Design (OED).

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The Multifaceted Mathematics for Predictive Digital twins (M2DT) collaboration addresses this challenge by developing a closed loop process by building up a hierarchy of interconnected models with varying levels of fidelity that can be updated with experimental data. One application within M2DT is the self-assembly of block copolymer thin films whose nanoscale morphologies determine important material properties relevant to semiconductor and battery manufacturing. High-fidelity Molecular Dynamics (MD) simulations accurately capture the self-assembly processes, but are prohibitively expensive for large parameter studies or optimal experimental design studies. To enable rapid exploration of processing conditions, reduced-order phase-field models based on the Cahn-Hilliard free energy functional are used as surrogates, meaning they are computationally more efficient models of the same system. [2] However, we don't know the parameters of this model, and how they are related to the actual BCP system, or to the MD model. Before the surrogate model can be used for predictive purposes and OED, the parameters must be calibrated. This work focuses on developing a Bayesian calibration framework for estimating uncertain Cahn-Hilliard model parameters from MD data. [1] Gaussian process machine learning models are used to emulate Cahn-Hilliard simulations, since although it is efficient, it still isn't efficient enough for calibration. Rather than calibrating to a single set of parameters for the Cahn-Hilliard surrogate, the Bayesian framework quantifies uncertainty and identifies the most plausible regions of parameter space consistent with observed MD data. This improves the predictive reliability of the Cahn-Hilliard model within the broader M2DT digital twin framework.

2 Methods

2.1 Molecular Dynamics Data Set

The observational data used for calibration were obtained from coarse-grained Molecular Dynamics Simulations of block copolymer thin-film self-assembly performed by the Center for Functional Nanomaterials (CFN) at Brookhaven National Laboratory using LAMMPS. These simulations model the evolution of diblock copolymer morphologies under prescribed interaction parameters. They resolve molecular-scale physics at substantially high fidelity than the surrogate Cahn-Hilliard model while remaining computationally tractable for generating calibration data.

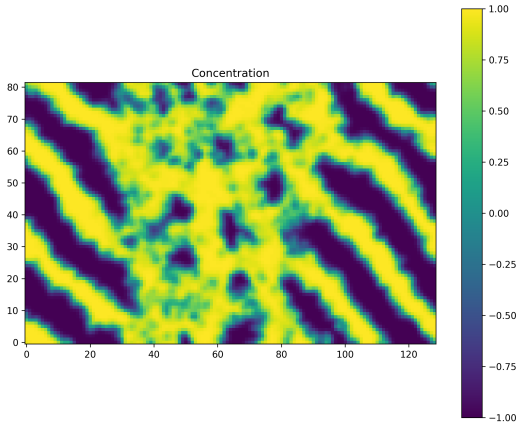


Figure 1: A snapshot of a molecular dynamics simulation

$$E[c] = \int_\Omega \left[\frac{1}{4}(c^2-1)^2 + \frac{\epsilon^2}{2}|\nabla c|^2\right] dV + \frac{\sigma}{2}\int_\Omega (c-m)(-\nabla^2)^{-1}(c-m)dV$$

Figure 2: Cahn Hilliard free energy functional with a non-local Ohta-Kawasaki term to account for long range interactions [2]

2.2 Cahn-Hilliard Surrogate Model

We model block copolymer self-assembly using a continuum phase-field description. The spatiotemporal phase field describes the evolution of the polymer concentration field. $c(x, y, z, t)$ denote the local composition of the two polymer species at position (x, y, z) and time t with prescribed spatial mean m. Unlike molecular dynamics which resolves individual particles, the phase field model represents the morphology evolving through time through this continuous concentration field. The dynamics are derived from a free energy functional containing a free energy double well potential, an interfacial penalty, and a long range interaction term. [2] The resulting Cahn-hilliard Ohta-Kawasaki equation is

$$\frac{dc}{dt} = \nabla^2 \left(\frac{\delta E}{\delta c} = \nabla^2 (c^{23} - c - \epsilon^2 c) - \frac{1}{\epsilon^2 \gamma^2}(c - m) \right)$$

Figure 3: Governing Evolution equation for the Cahn-Hilliard surrogate model. A fourth-order partial differential equation.

Where ϵ^2 controls the energetic penalty associated with sharp interfaces and sigma controls the strength of the long range interactions of the polymers. In the parameterization used in this work, $\sigma = \frac{1}{\epsilon^2 \gamma^2}$, where γ is an interaction length parameter.

The Cahn-Hilliard model is used as a physics based surrogate for the MD simulations. It retains the mechanisms responsible for the evolution of the system, specifically the energetic penalty of sharp interfaces, the double well free energy potential, and the long range interaction of polymers, but without the computational cost of explicitly resolving polymer chains. The calibration problem is therefore to infer values of m, ϵ, and γ for which the surrogate produces Quantities of interest consistent with those extracted from the MD data.

2.3 Fast Fourier Transform (FFT) Spectral Solver

We compute the numerical solutions of the governing equation using a first order semi-implicit spectral method implemented in C++. The computational domain is periodic in the x and y directions and satisfies homogeneous Neumann boundary conditions in the z direction. Accordingly, Fast Fourier Transforms (FFTs) are applied in the periodic directions, and a Discrete Cosine transform (DCT) is used in z to satisfy the Neumann boundary conditions.

A semi-implicit time discretization is a better choice than fully explicit methods, because the fourth-order biharmonic operator is numerically stiff. Transforming the governing equation into spectral space diagonalizes the linear differential operators, reducing each timestep to efficient algebraic operations while preserving numerical stability.

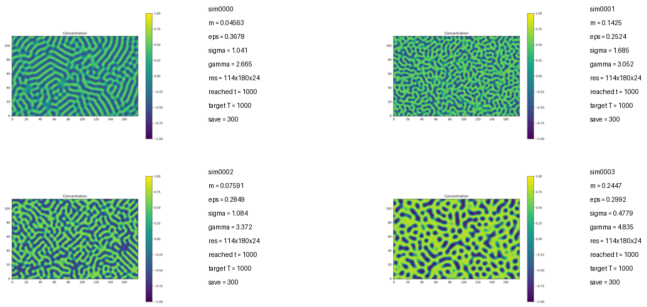


Figure 4: Cahn-Hilliard simulations

2.3.1 Timescales

We used adaptive time stepping that hovered around $dt = 1e-2$ and $1e-3$, which was a significant improvement from the backwards difference methods (BDF) with multigrid solvers and fourth order Runge-Kutta solvers that were previously implemented, which required time steps around $dt = 1e-6$ to $dt = 1e-9$ respectively, and at which the solve is more computationally expensive at each step. It might take the Runge-Kutta solver around 16 hours to reach a final time of 1.5, the BDF multigrid solver 7 hours to reach a final time of 10, and the semi-implicit spectral solver less than 3 hours to reach a final time of 1,000. This was important because the model has different approximated timescales in different areas of parameter space. Parameter sets previously believed to produce negligible evolution had simply not been simulated long enough. When evolved for much longer time, these simulations developed distinct morphologies. It is also important to let the simulation get to an equilibrium state with minimal oscillations. Since we extract Quantities of Interest only from the final state, it is important to integrate long enough for transient oscillations in the system to decay for reliable calibration data.

The characteristic evolution time varies substantially across parameter space, making the ability to efficiently integrate over long physical times particularly important. Spectral methods are also particularly suitable for phase-field models because the concentration field is smooth. It provides rapid convergence while accurately resolving the interfacial structure.

2.4 Quantities of Interest

Although the molecular dynamics and Cahn-Hilliard simulations are modeling the same system, when comparing large amounts, we need an efficient and effective way to compare two differently

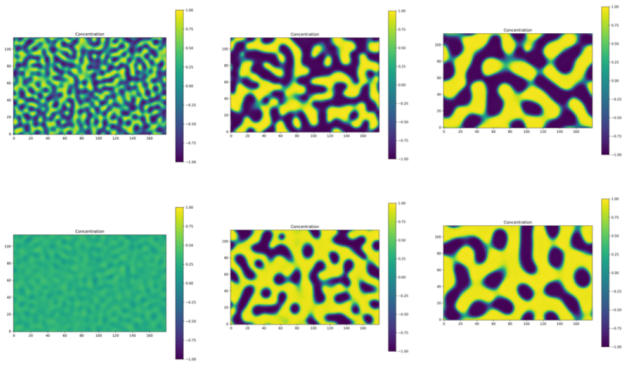


Figure 5: Snapshots of 2 simulations at different times. Column 1: Time = 10. Column 2: Time = 100. Column 3: Time = 1,000

parametrized systems, where one is an approximation of the other. Instead, we map each 3D concentration field into a scalar quantities of interest (QoIs) that capture the dominant structural characteristics while providing a common representation for comparing MD and CH simulations. More importantly, we care about the major structures and features in the morphologies, not the specific pixel concentrations. To capture these main features, we map the domain to a 2D power spectrum using a Fourier transform. This is mapped to a 1D peak, from which we gather scalar values from the location, width, and amplitude of this peak. These 3 QoIs give us information about the characteristic length scale throughout the morphology, spacing between domains, and how organized and repeating the morphology is. We also observe the angular distribution of the Fourier spectrum to get our fourth QoI, which captures information about how well ordered it is in six directions, also called anisotropy. These provide a common space to directly compare Molecular dynamics and Cahn-Hilliard model outputs to calculate residual for the Cahn-Hilliard model.

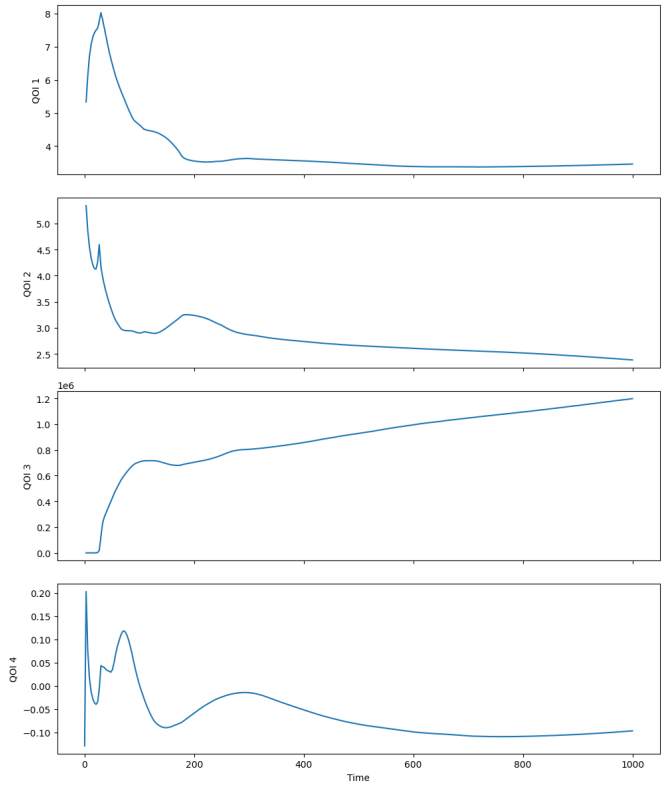


Figure 6: As the concentration field evolves, the QoIs are calculated for each snapshot. Convergence is desirable.

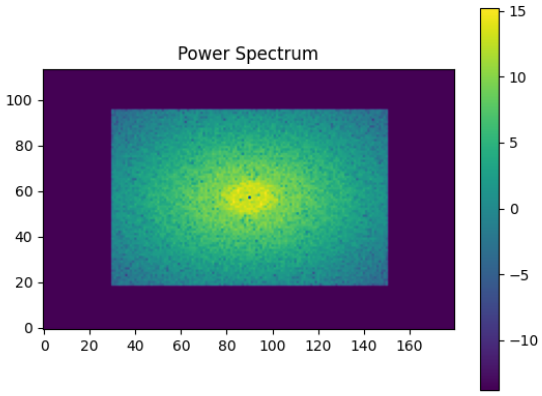


Figure 7: Radially averaged power spectrum, from which QoIs 1,2,3 are derived

2.5 Gaussian Process Model

Bayesian calibration requires evaluating the forward model thousands of times to explore the posterior distribution. Although the Cahn-Hilliard model is substantially less expensive than Molecular Dynamics, repeatedly solving the governing partial differential equation during Markov Chain monte Carlo is still computationally prohibitive for thousands or tens of thousands of samples. To make this feasible, we train a Gaussian Process (GP) emulator to learn the mapping of parameters to their QoIs that would be produced by the Cahn-Hilliard and QoI workflow. We train a separate GP emulator on each QoI in the same way. Predictive accuracy was assessed using leave-one-out cross validation of the four QoI emulators to estimate emulator error over the training ensemble. Unlike deterministic interpolation methods, the GP emulator provides both a predictive mean and an estimate of predictive uncertainty. This uncertainty is propagated into the Bayesian likelihood to account for emulator error during calibration. During calibration, the emulator replaces repeated PDE solves while providing an estimate of the uncertainty associated with each prediction. The total observation variance is modeled as

$$\sigma^2_{total} = \sigma^2_{GP} + \sigma^2_{RMS} + \sigma^2_{reg}$$

where σ^2_{GP} is the GP predictive variance, σ^2_{RMS} is the leave-one-out RMS emulator error, and σ^2_{reg} is a small numerical regularization term. Incorporating emulator uncertainty allows Bayesian inference to account for both interpolation error and finite training data.

2.6 Bayesian Calibration

The calibration problem is formulated as a Bayesian inverse problem, where the objective is to infer the Cahn-Hilliard model paramteres that produce morphologies consistent with the Molecular Dynamics observations. Rather than identifying a single best-fit parameter set, Bayesian calibration estimates a posterior probability distribution ove the parameter space, thereby quantifying uncertainty arising from limited observations, surrogate error, and model discrepancy. Uniform prior distributions were assigned over parameters m, ϵ, γ that produced coherent morphologies in the Cahn-Hilliard model. Assuming independent Gaussian observation errors, the likelihood is constructed from the residual between the predicted and observed QoIs using the total variance

described in Section 2.5. the posterior distribution is obtained through Bayes' Theorem.

$$p(\theta|y) \propto p(y|\theta)p(\theta)$$

where $p(\theta)$ denotes the prior distribution $p(y|\theta)$ is the likelihood, and $p(\theta|y)$ is the posterior distribution. The posterior therefore identifies regions of parameter space that are most consistent with the observed Molecular Dynamics morphologies while accounting for uncertainty in both the surrogate model and the emulator. Because the posterior distribution cannot be evaluated analytically, samples were generated using the No-U-Turn Sampler, an adaptive Hamiltonian Monte Carlo algorithm that efficient explores continuous parameter spaces without manual tuning of proposal distributions. Multiple independent Markov chains were run. We ran standard convergence diagnostics, including trace plots, the potential scale reduction and factor $\hat{R}$ effective sample size to diagnose runs.

3 Results

3.1 Gaussian Process Validation

Before performing Bayesian calibration, the Gaussian Process emulator was evaluated using leave-one-out cross validation. Root mean squared error (RMSE) was computed for each quantity of interest to verify that the surrogate accurately reproduced the Cahn-Hilliard simulations. The low predictions errors indicate that the GP emulator provides an accurate approximation of the forward model while reducing the computational cost of evaluating the likelihood during MCMC.

3.2 Markov Chain Monte Carlo with No-U-Turn sampling algorithm

Hamiltonian Monte Carlo using the No-U-Turn Sampler successfully converged for each calibration problem. five independent chains were run with 1,500 warmup iterations, and 5,000 posterior samples per chain. Average acceptance probabilities ranged around 0.80 to 0.90, indicating stable sampling performance by the algorithm. Tree depths remained relatively small, which indicates that the posterior was expored efficiently without requiring excessively long trajectories. The likelihood covariance reports confirmed that the likelihood was modeled using a diagonal covariance matrix, consistent with the asusmption that the QoI observation errors are independent.

3.3 Posterior distributions conditioned on individual QoIs

The posterior distributions obtained by conditioning on each QoI separately demonstrate that the four QoIs provide different amounts of information and sensitivity to the model parameters.

3.4 3.3 Molecular Dynamics Targets

Posterior distributions obtained by calibrating to a single MD simulation were compared with those obtained by calibrating to the average of all MD simulations. The posterior shapes were qualitatively similar between the two calibration targets. Major modes remained in the same parameter regions, and the sensitivity of QoIs were largely the same. Averaging the MD simulations reduced random variability in the target observations, but didn't change the inferred parameter distirbutions, indicating that the Bayesian calirbation framework is reasonably robust to the choice of MD target and the dominant source of uncertainty arises from parameter identifiability rather than stochastic variation among MD simulations.

3.5 Parameters

The strongest parameter constraints were observed for ϵ^2. Conditioning on each QoI showed that smaller values of ϵ^2 were favored, however QoI 4 was the most sensitive and provided the most information. γ largely resembled the uniform prior, indicating that it might be as relevant to a change in morphology, but it is more likely that the QoIs are not sensitive to a change in the γ parameter. This may be because γ seemed to affect the timescale of the system rather than the morphology. Initially, tests were run for average concentration m on the range [-1,1], and a bimodal distribution was consistently sampled, favoring higher absolute values of m. Recall that in the Cahn-Hilliard equation, mass is conserved, so this value is constant throughout the evolution. Then we constrained the sampling range to [0,1], which shouldn't change the distribution since m is average concentration of two phases, so it doesn't matter whether the phase is A or B, because they are treated equally by the simulation. We found that the conditioning on the different QoIs produced different results on the favored values of m, although notably QoI 4 favored values of m close to 0 more heavily, while the other QoIs produced broadly similar distributions. These results suggest that the available QoIs provide only weak information regarding m when considered

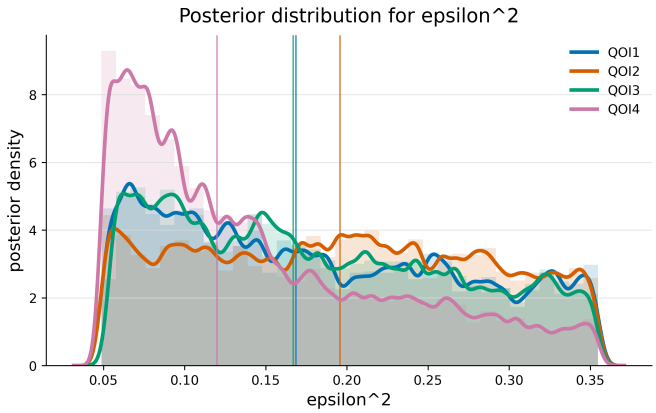


Figure 8: Posterior distribution over ϵ^2

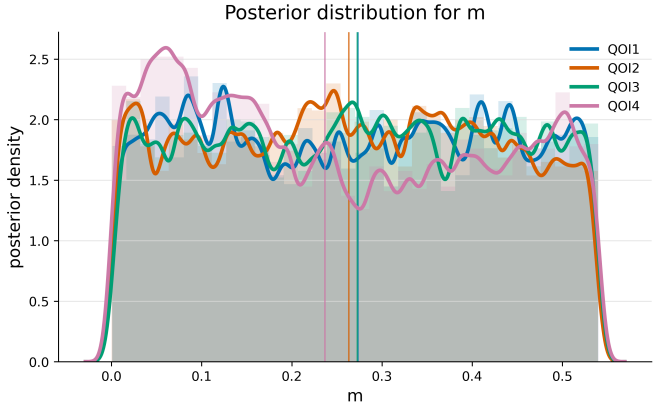


Figure 9: Posterior distribution over m

individually.

3.6 Timescale and evolution

We found that the physical rate of evolution depended on several things. The ϵ parameter has a strong influence on the timescale, because it controls the interfacial width, which is the boundaries between distinct domains. In the Cahn-Hilliard partial differential equation, ϵ is on the fourth order biharmonic term/gradient term, so a small ϵ results in larger gradients and thinner interfaces. this often produces faster interfacial relaxation, in other words, the system reaches a stable state faster. The γ parameter is related to $\sigma = \frac{1}{\epsilon^2\gamma^2}$ which is the coefficient of the nonlocal Ohta-

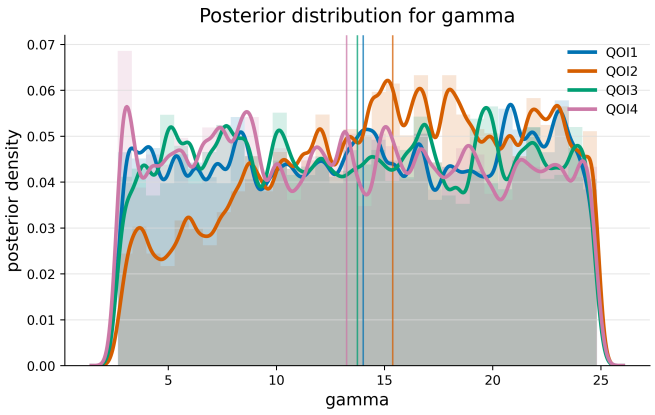


Figure 10: Posterior distribution over γ

Kawasaki term to account for the long range interactions of the BCP. Since γ is inversely related to σ, a large γ or smaller σ is associated with weaker long-range interactions, which in turn results in slower large-scale ordering. In the same way, smaller γ causes stronger long-range interactions, which results in faster domain formation. We used random initial conditions to allow the system to undergo spinodal decomposition. If the simulation starts already phase separated, then it may get trapped in an energy well and evolve towards a final state artificially. Adaptive time stepping is advantageous because the system evolves very quickly in the beginning, so a smaller timestep better captures this evolution. As the system's evolution slows down closer and approaches a stable state, the timestep is allowed to increase according to an error threshold.

4 Discussion

The results demonstrate that Bayesian calibration provides considerably more information than a single optimal parameter estimate. The posterior distributions quantify which regions of parameter space remain consistent with the Molecular Dynamics observations and reveal the relative identifiability of each parameter. Although posterior distributions were broadly consistent across different MD targets, some variability remained, which indicate uncertainty introduced in each step of the calibration workflow could propagate in the final posterior distributions. One area of further exploration might be finding a better way of computing the residual between the Cahn-Hilliard surrogate model and the calibration data. In other words, finding better Quantities of Interests that are the most sensitive to parameters, therefore they are the most informative when

finding posterior distributions. This will help to better distinguish informative regions of parameter space. Although posterior distributions across MD targets largely resembled each other, there is still variation, which tells us that the MD simulations aren't consistent either. Another area of further exploration might be exploring how the noise term can be calculated in the likelihood function. There is debate whether including certain noise terms produce more accurate results.

5 Acknowledgements

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